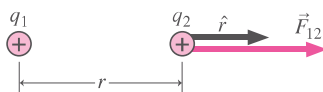


Big Picture

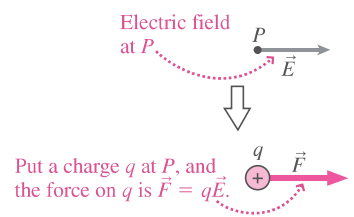
This chapter introduces several big ideas. First is **electric charge**, a fundamental property of matter that comes in positive and negative forms. Like charges repel and opposites attract; this is the **electric force**. It's convenient to define the **electric field** as the force per unit charge that a charge would experience if placed in the vicinity of other charges. Both force and field obey the **superposition principle**, meaning that the effects of several charges add vectorially.

Key Concepts and Equations

Coulomb's law describes the electric force between point charges:

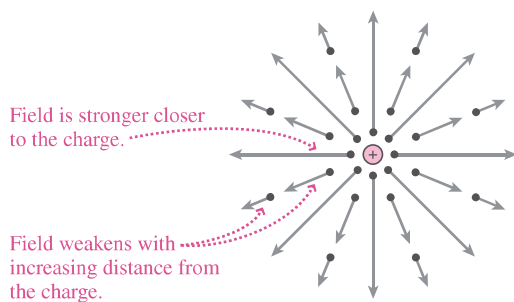
$$\vec{F}_{12} = \frac{kq_1q_2}{r^2} \hat{r}$$


The electric field is the force per unit charge, $\vec{E} = \vec{F}/q$, and therefore the force a given charge q experiences in a field is $\vec{F} = q\vec{E}$.

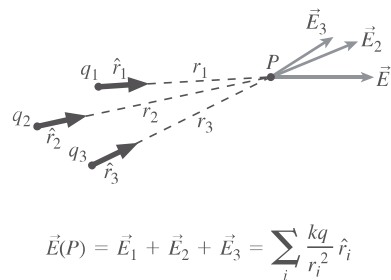


The field of a point charge follows from Coulomb's law:

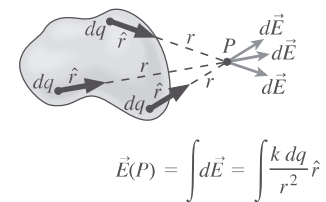
$$\vec{E} = \frac{kq}{r^2} \hat{r}$$



Fields of charge distributions are found by summing fields of individual point charges, or by integrating in the case of continuously distributed charge:



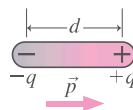
$$\vec{E}(P) = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \sum_i \frac{kq}{r_i^2} \hat{r}_i$$



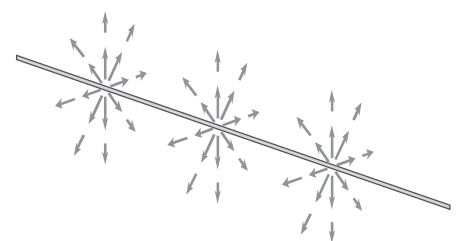
$$\vec{E}(P) = \int d\vec{E} = \int \frac{k dq}{r^2} \hat{r}$$

Applications

A **dipole** consists of equal but opposite charges $\pm q$ a distance d apart. For distances large compared with d , the dipole drops as $1/r^3$, and the dipole is completely characterized by its **dipole moment** $\vec{p} = q\vec{d}$.

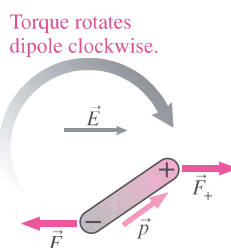


The field of an infinite line drops as $1/r$: $E = 2k\lambda/r$, with λ the charge per unit length. This is a good approximation to the field near an elongated structure like a wire.



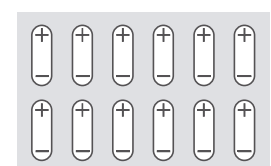
Point charges respond to electric fields with acceleration proportional to the charge-to-mass ratio q/m .

A dipole in an electric field experiences a torque that tends to align it with the field: $\vec{\tau} = \vec{p} \times \vec{E}$.



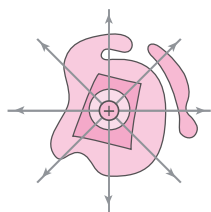
If the field is nonuniform, there's also a net force on the dipole.

Dielectrics are insulating materials whose molecules act like electric dipoles.



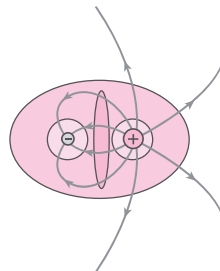
Big Picture

The big idea here is **Gauss's law**—a universal statement about electric fields that's closely related to Coulomb's inverse-square law but expressed in terms of the global behavior of the field over any closed surface. Using the **electric field-line** picture, Gauss's law says that the number of field lines emerging from a closed surface depends only on the net charge enclosed; more rigorously, it says that the **electric flux** through the surface is proportional to the enclosed charge.



Point charge q

Eight lines pass through any closed surface surrounding q .



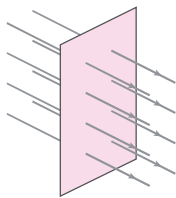
Point charges $+q, -q/2$

The number of lines through a surface depends on the net charge enclosed.

Key Concepts and Equations

Electric flux Φ describes the amount of electric field crossing an area.

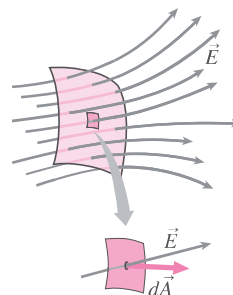
$\Phi = EA$ for a flat area perpendicular to a uniform field



In terms of flux, Gauss's law reads $\oint \vec{E} \cdot d\vec{A} = q_{\text{enclosed}}/\epsilon_0$.

Here $\epsilon_0 = 1/4\pi k = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$ is another way of expressing the Coulomb constant $k = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.

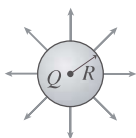
$$\Phi = \int \vec{E} \cdot d\vec{A}, \text{ in general}$$



Applications

Gauss's law gives the fields of symmetric charge distributions:

Spherical symmetry:



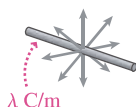
$$\text{Outside: } E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{kQ}{r^2}$$

Inside uniformly charged sphere:

$$E = \frac{kQr}{R^3}$$

Inside hollow sphere: $E = 0$

Line symmetry:

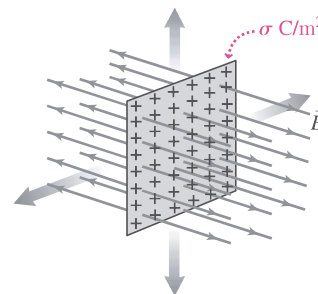


$$\text{Outside: } E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Inside hollow pipe:

$$E = 0$$

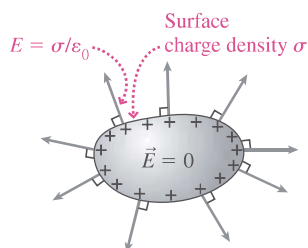
Plane symmetry:



$$\text{Outside charged slab: } E = \frac{\sigma}{2\epsilon_0}$$

Gauss's law and conductors:

- The field is zero inside a conductor in electrostatic equilibrium.
- Any net charge resides on a conductor's surface.
- The field at the surface is perpendicular to the surface and has magnitude σ/ϵ_0 .



Big Picture

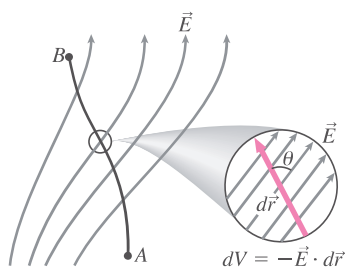
The big idea here is **electric potential difference**—a measure of the energy per unit charge involved in moving charge between two points in an electric field. Because the electric field is conservative, potential difference is path independent and thus depends only on the two points in question.

Key Concepts and Equations

Electric potential difference between points A and B is the negative of the line integral of the electric field over any path from A to B :

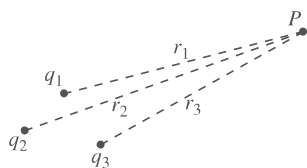
$$\Delta V_{AB} = \frac{\Delta U_{AB}}{q} = - \int_A^B \vec{E} \cdot d\vec{r}$$

When a charge “falls” through a potential difference ΔV , it gains energy $q\Delta V$.

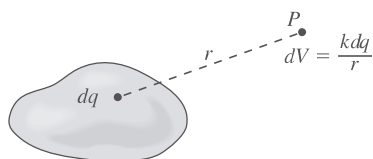


Potentials of charge distributions follow by summing or integrating the fields of pointlike charge elements:

$$V = \sum \frac{kq_i}{r_i} \quad (\text{discrete charges})$$

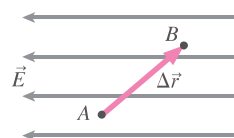


$$V = \int \frac{k dq}{r} \quad (\text{continuous charge distribution})$$

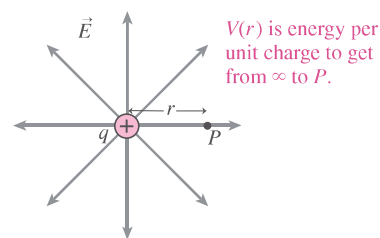


In a uniform field, the potential difference becomes

$$\Delta V_{AB} = -\vec{E} \cdot \Delta\vec{r}$$

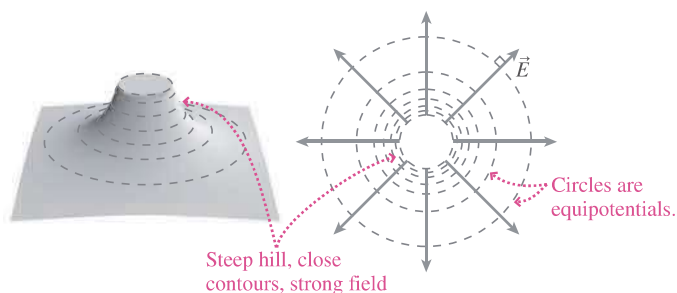


The potential in the field of a point charge is $V(r) = kq/r$, where the zero of potential is taken at infinity and r is the distance from the point charge.



Equipotentials are surfaces of constant potential perpendicular to the electric field. Where equipotentials are close, the field is strong. The field component in a given direction depends on the rate at which potential changes with position; thus,

$$E_x = -\frac{dV}{dx}$$

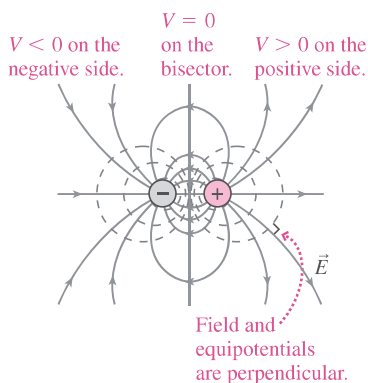


Applications

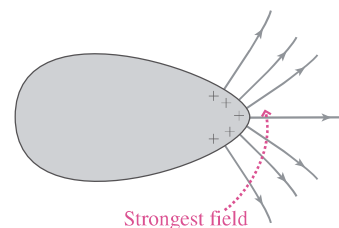
The dipole potential is

$$V = \frac{kp \cos \theta}{r^2}$$

where $p = qd$ is the dipole moment and the angle θ is measured from the dipole axis.



In charged conductors, the charge density is generally highest, and the field strongest, where a conductor curves sharply.

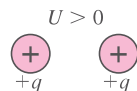


Big Picture

The big idea here is that *all* electric fields represent stored energy. This energy is associated with the work needed to assemble a distribution of electric charge, and may be negative or positive.

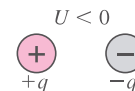
You do positive work to assemble this charge distribution . . .

. . . and therefore the stored electric energy U is positive.



You do negative work to assemble this charge distribution . . .

. . . and therefore the stored electric energy U is negative.



Key Concepts and Equations

The **energy density** in an electric field E is $u_E = \frac{1}{2}\epsilon_0 E^2$.

Integrating over volume gives the total electric energy U stored in the field:

$$U = \int u_E dV.$$

At this point the field strength is E , so there's electric energy whose density is $u_E = \frac{1}{2}\epsilon_0 E^2$.



The electric energy U stored in the shaded volume is

$$U = \int u_E dV = \frac{1}{2}\epsilon_0 \int E^2 dV.$$

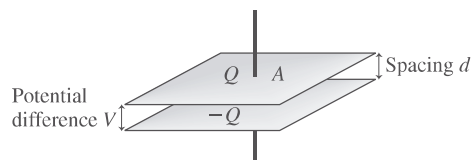
Applications

A **capacitor** is a pair of insulated conductors used to store electric energy. **Capacitance** is the ratio of charge to potential difference:

$$C = Q/V$$

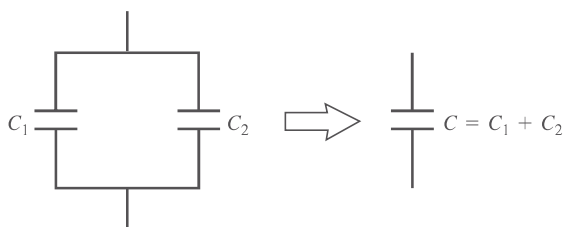
For a parallel-plate capacitor:

$$C = \epsilon_0 A/d$$



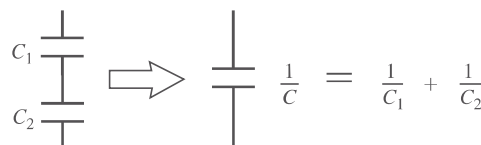
The energy stored in a capacitor is $U = \frac{1}{2}CV^2$.

Capacitors in parallel add: $C = C_1 + C_2$.



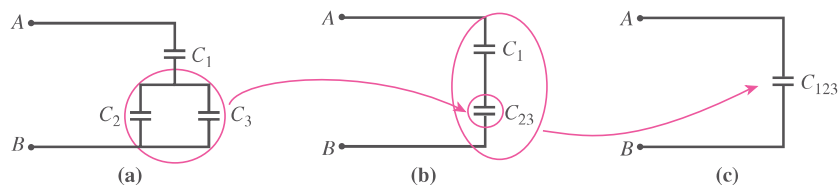
Capacitors in parallel have the same voltage.

Capacitors in series add reciprocally: $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$.



Capacitors in series have the same charge.

Complicated circuits are analyzed by breaking them into parallel and series combinations:



A **dielectric** between capacitor plates increases the capacitance, as determined by the **dielectric constant** κ of the material: $C \rightarrow \kappa C_0$.

Big Picture

The big idea here is **electric current**—the flow of electric charge—and its microscopic cousin, **current density**. With current we don't have electrostatic equilibrium, and there's usually an electric field in a current-carrying conductor. **Ohm's law** is an empirical statement—not a fundamental law of physics—that relates current and voltage, or current density and electric field.

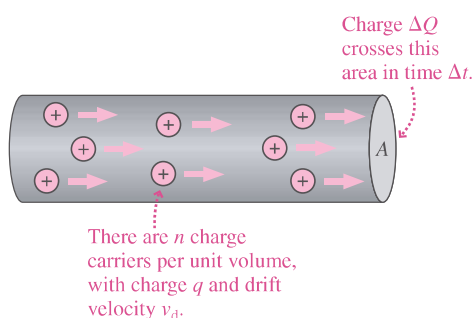
Key Concepts and Equations

Quantitatively, **current** is defined as the rate of charge flow:

$$I = \frac{\Delta Q}{\Delta t}$$

Current density is the current per unit area. Its magnitude is

$$J = \frac{I}{A}$$



Microscopically, current depends on the density of charge carriers, their charge, and the **drift velocity**:

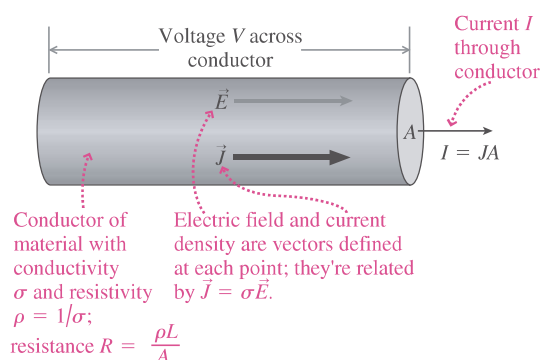
$$I = nqAv_d \quad \text{and} \quad \vec{J} = nq\vec{v}_d$$

The microscopic version of **Ohm's law** relates electric field, current density, and **conductivity** σ (or its inverse, **resistivity** ρ):

$$\vec{J} = \sigma \vec{E}$$

The macroscopic version relates voltage, current, and resistance:

$$I = V/R$$



Electric power is the product of voltage and current:

$$P = IV$$

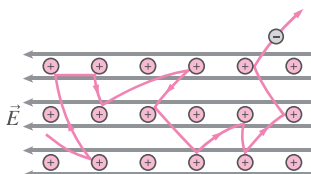
Using Ohm's law, this can also be written

$$P = I^2 R$$

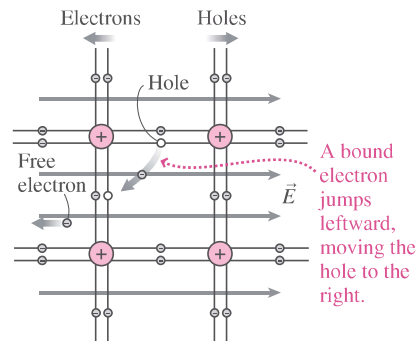
$$P = \frac{V^2}{R}$$

Applications

Different types of conductors have different conduction mechanisms. In **metals**, free electrons carry the current; in **ionic solutions**, both positive and negative ions are involved; in **plasmas**, the charge carriers are free electrons and ions; and in **semiconductors**, both electrons and positive holes carry current, with semiconductor conduction properties readily adjustable. **Superconductors** are materials that exhibit zero resistance at sufficiently low temperatures.



Electron and hole move oppositely in an electric field.



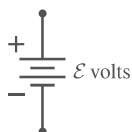
Electrical safety is a matter of avoiding currents high enough to cause biological harm, and that means avoiding voltages high enough to drive such currents.

Big Picture

The big idea here is the **electric circuit**—an interconnection of electric components that usually includes one or more sources of electric energy, such as batteries.

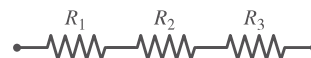
Key Concepts and Equations

A source of **emf** (or an emf, for short) is a battery or other device that imparts energy to electric charge flowing through it. The value of the emf, $\mathcal{E}$, is the energy imparted per unit charge, measured in volts. An ideal emf maintains a fixed potential difference (voltage) across its terminals.



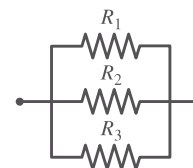
Resistors in series add:

$$R_{\text{series}} = R_1 + R_2 + R_3 + \dots$$

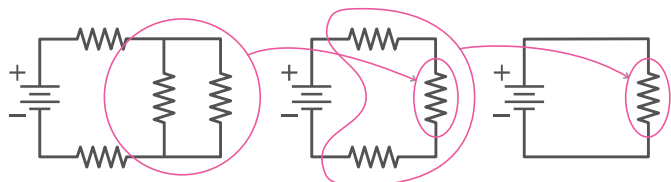


Resistors in parallel add reciprocally:

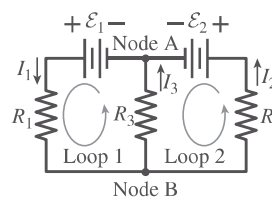
$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$



Simple circuits are analyzed by evaluating series and parallel combinations.



To analyze more complicated circuits, use Kirchhoff's node and loop laws.



Current into node A is 0:

$$-I_1 + I_2 + I_3 = 0$$

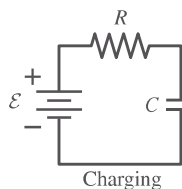
Voltage changes around loops sum to 0:

$$\text{loop 1: } \mathcal{E}_1 - I_1 R_1 - I_3 R_3 = 0$$

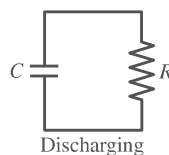
$$\text{loop 2: } \mathcal{E}_2 + I_2 R_2 - I_3 R_3 = 0$$

Capacitors result in time-changing behavior of circuit quantities.

The **time constant** RC governs the time scales.



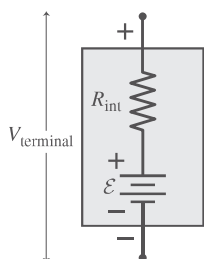
$$V_C = \mathcal{E}(1 - e^{-t/RC})$$



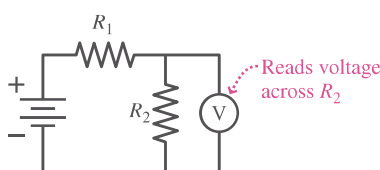
$$V_C = V_0 e^{-t/RC}$$

Applications

Batteries and other electric-energy sources have **internal resistance**. When they supply current, their terminal voltage is therefore less than their rated voltage $\mathcal{E}$.

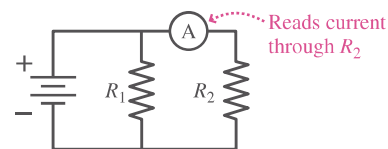


A **voltmeter** measures the voltage *across* its two terminals. It goes in parallel with the component whose voltage you want to measure.



An ideal voltmeter has infinite resistance.

An **ammeter** measures the current *through* itself. It goes in series with the component whose current you want to measure.



An ideal ammeter has zero resistance.

CHAPTER 26

SUMMARY

Big Picture

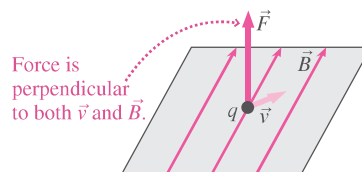
The big new idea here is magnetism—an interaction that fundamentally involves *moving electric charge*. Moving charge produces magnetic fields, and moving charges respond to magnetic fields by experiencing a magnetic force.

Key Concepts and Equations

The **magnetic force** on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is

$$\vec{F} = q\vec{v} \times \vec{B} \quad (\text{magnetic force})$$

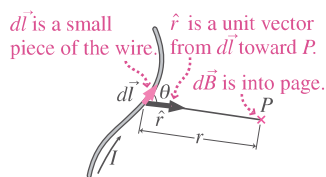
The force acts at right angles to both $\vec{v}$ and $\vec{B}$, and therefore it does no work.



The **Biot-Savart law** describes the magnetic field $d\vec{B}$ arising from a small element of steady current:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Here μ_0 is the **permeability constant**, with value $4\pi \times 10^{-7} \text{ N/A}^2$.



Ampère's law provides a more global description of how magnetic fields arise from currents, relating the line integral around any closed loop to the encircled current:

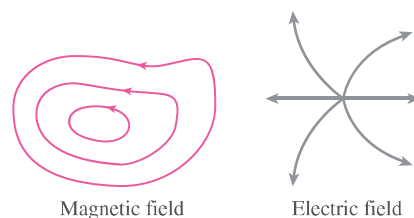
$$\oint \vec{B} \cdot d\vec{r} = \mu_0 I_{\text{encircled}}$$

Ampère's law in this form applies only to steady currents.

Gauss's law for magnetism expresses the fact that there are no magnetic monopoles—magnetic analogs of electric charge—and that magnetic field lines therefore do not begin or end:

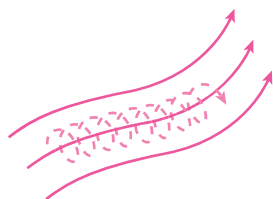
$$\oint \vec{B} \cdot d\vec{A} = 0$$

Static electric fields, in contrast, always begin or end on electric charges.

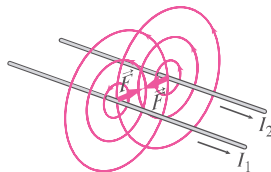


Applications

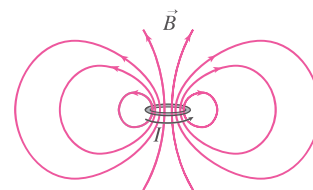
A charged particle moving perpendicular to a uniform magnetic field undergoes circular motion with the **cyclotron frequency** $f = qB/2\pi m$. More generally, charged particles in magnetic fields follow spiral paths, “trapped” on the field lines.



The magnetic force on a straight wire of length l carrying current I in a uniform magnetic field is $\vec{F} = I\vec{l} \times \vec{B}$. Parallel wires a distance d apart experience forces from each other's magnetic field: $F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$. The force is attractive for currents in the same direction, repulsive for currents in opposite directions.



A current loop gives rise to a magnetic field that, at distances large compared with the loop's size, is a dipole field. The loop's magnetic dipole moment has magnitude $\mu = IA$, with A the loop area, and the loop responds to an external magnetic field by experiencing the torque typical of a dipole: $\vec{\tau} = \vec{\mu} \times \vec{B}$.

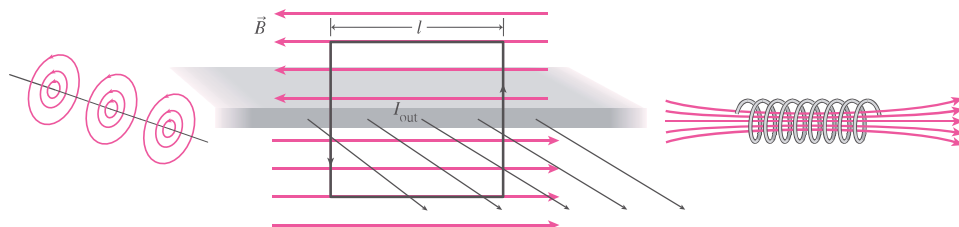


Fields of simple current distributions:

Line current: $B = \frac{\mu_0 I}{2\pi r}$

Current sheet: $B = \frac{1}{2} \mu_0 J_s$

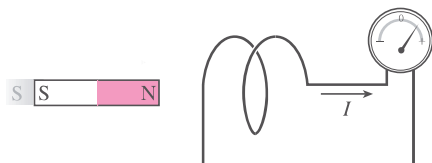
Solenoid: $B = \mu_0 nI$



Magnetism in matter arises from the interactions of atomic-scale current loops. **Ferromagnetic** materials have strong interactions and exhibit the bulk magnetism associated with permanent magnets and with magnetic materials like iron. **Paramagnetism** and **diamagnetism** are weaker manifestations of magnetism in matter.

Big Picture

The big idea here is **electromagnetic induction**, a phenomenon in which a **changing magnetic field produces an electric field**. Applied to circuits, induction results in induced emfs that drive induced currents.



Here a moving magnet produces the changing magnetic field.

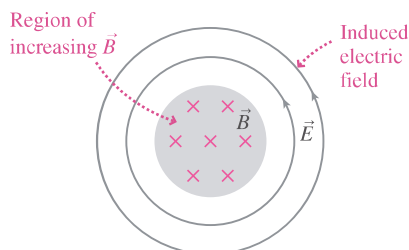


Here a change in current produces the changing magnetic field.

Key Concepts and Equations

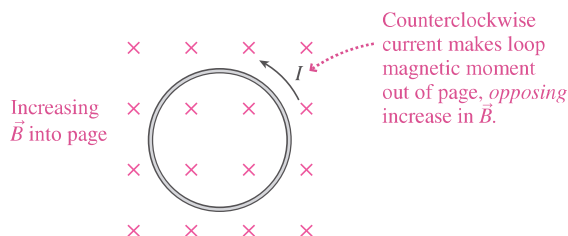
Faraday's law describes induction quantitatively, relating the line integral of the induced electric field to changing magnetic flux:

$$\oint \vec{E} \cdot d\vec{r} = -d\Phi_B/dt$$



In conductors, Faraday's law gives the induced emf: $\mathcal{E} = -d\Phi_B/dt$.

Lenz's law shows that electromagnetic induction is consistent with conservation of energy, which requires that induced effects act to oppose the changes that give rise to them.

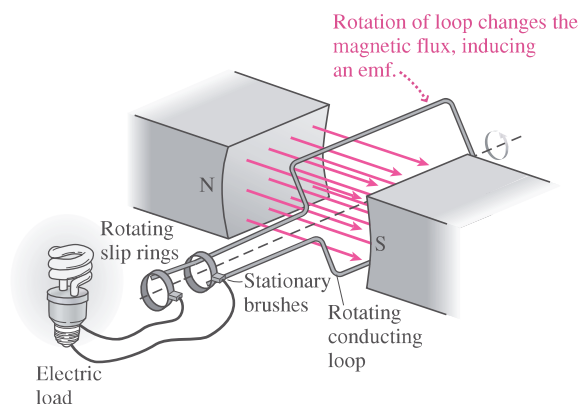


Magnetic fields contain stored energy, as do electric fields. The **magnetic-energy density** is

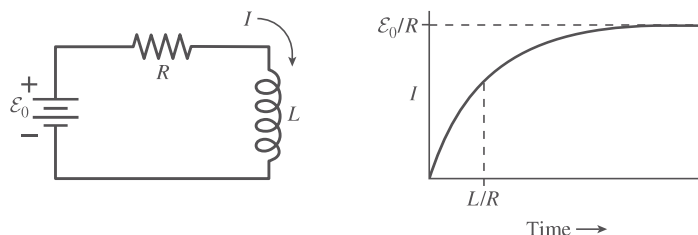
$$u_B = \frac{B^2}{2\mu_0}$$

Applications

Electric generators convert mechanical to electrical energy by moving conductors in magnetic fields to induce emfs that drive currents.



Inductors are wire coils that encircle their own magnetic flux, giving **self-inductance** $L = \Phi_B/I$. An inductor opposes changes in current, producing an emf given by $\mathcal{E} = -L(dI/dt)$. Circuit quantities in a simple RL circuit change with **inductive time constant** L/R .



$$I = \frac{\mathcal{E}_0}{R} (1 - e^{-Rt/L})$$

Diamagnetism occurs when electromagnetic induction results in atoms acquiring net magnetic moments; the result is a repulsive interaction.

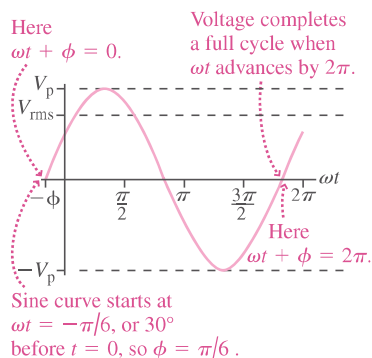
Big Picture

The big idea here is **alternating current (AC)**, which in its simplest form exhibits sinusoidal variation in current and voltage. Resistors respond to AC as to DC, with current directly proportional to voltage. In capacitors and inductors, the current–voltage relation depends on frequency, and the current and voltage are out of phase.

Key Concepts and Equations

An AC voltage or current is characterized by its amplitude (rms or peak value), its frequency, and phase:

$$V = V_p \sin(\omega t + \phi)$$



Reactance, X , characterizes the relation between peak (or rms) current and voltage in capacitors and inductors:

$$I_p = \frac{V_p}{X}$$

Capacitor:

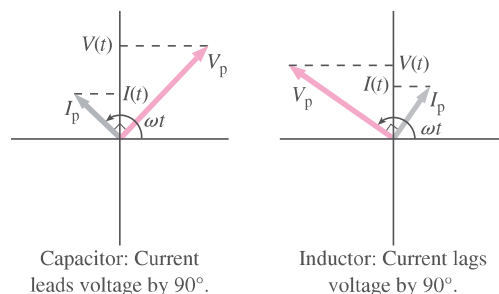
$$X_C = 1/\omega C$$

Inductor:

$$X_L = \omega L$$

where $\omega = 2\pi f$ is the angular frequency of the AC voltage and current.

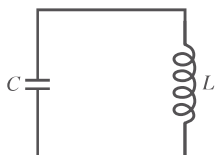
Phasors are arrows used to describe time-varying AC quantities. They rotate with angular velocity equal to the angular frequency ω .



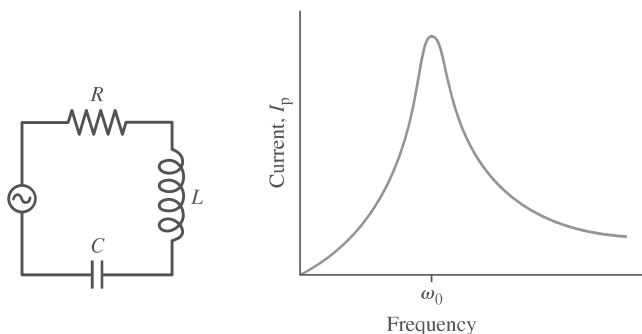
Applications

In an **LC circuit**, energy oscillates between electric and magnetic forms with frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



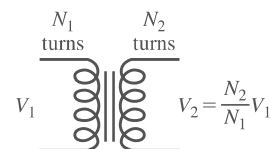
In a series **RLC circuit**, capacitor and inductor voltages cancel at the **resonant frequency**, ω_0 . Here the circuit exhibits the minimum **impedance**, $Z = \sqrt{R^2 + (X_L - X_C)^2}$, and passes the maximum current. The phase difference between voltage and current is $\tan \phi = (X_L - X_C)/R$.



The average power in an AC circuit depends on the cosine of the phase difference, also called the **power factor**:

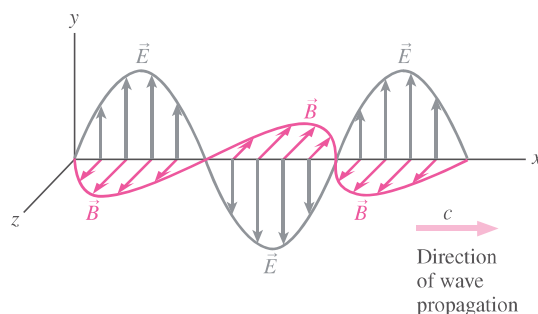
$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} \cos \phi$$

Transformers use electromagnetic induction to change voltage levels, transferring electric power between two circuits. **Diodes** and capacitive filters change AC to DC.



Big Picture

The big idea here—and one of the biggest ideas in physics—is that electric and magnetic fields together form self-regenerating structures that propagate through space as **electromagnetic waves**. What makes these waves possible is that changing magnetic fields induce electric fields (Faraday's law), and changing electric fields induce magnetic fields (Ampère's law, with Maxwell's modification). Electromagnetic (EM) waves in vacuum consist of electric and magnetic fields perpendicular to each other and to the direction of wave propagation, and in phase.



Key Concepts and Equations

Maxwell's equations describe completely the behavior of electric and magnetic fields in classical physics:

Law	Mathematical Statement	What It Says
Gauss for $\vec{E}$	$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$	How charges produce electric fields; field lines begin and end on charges.
Gauss for $\vec{B}$	$\oint \vec{B} \cdot d\vec{A} = 0$	No magnetic charge; magnetic field lines don't begin or end.
Faraday	$\oint \vec{E} \cdot d\vec{r} = -\frac{d\Phi_B}{dt}$	Changing magnetic flux produces electric fields.
Ampère	$\oint \vec{B} \cdot d\vec{r} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$	Electric current and changing electric flux produce magnetic fields.

Maxwell's equations show that electromagnetic waves are possible and that their speed in vacuum, the speed of light c , is related to the electric and magnetic constants ϵ_0 and μ_0 :

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

The value of c is very nearly 3.00×10^8 m/s. Its exact value, used in defining the meter, is 299,792,458 m/s.

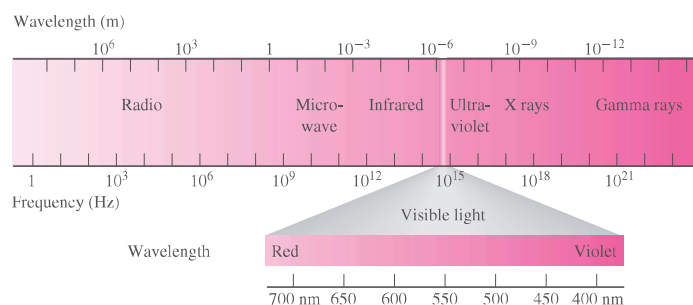
In vacuum, the electric and magnetic fields of a wave are related by

$$E = cB$$

The wave's frequency and wavelength are related by

$$f\lambda = c$$

EM waves can have any wavelength; the whole range constitutes the **electromagnetic spectrum**.



Applications

Polarization describes the direction of an EM wave's electric field and is a property widely used in scientific research and in technological devices including the ubiquitous liquid crystal displays. When polarized light of intensity S_0 is incident on a polarizer with its transmission axis at angle θ to the polarization, the light emerges with intensity

$$S = S_0 \cos^2 \theta$$

EM waves carry both energy and momentum. The **Poynting vector**

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

describes the rate of energy flow per unit area, while the momentum flow results in a **radiation pressure**:

$$P_{\text{rad}} = \frac{\bar{S}}{c}$$

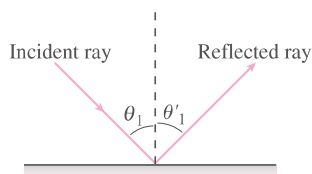
Big Picture

The big idea here is that light can be considered to travel in straight **rays** when the objects with which it interacts are much larger than the wavelength. Under these conditions, light rays **reflect** and **refract** at interfaces between different materials.

Key Concepts and Equations

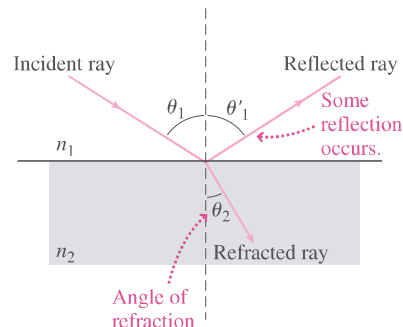
The **angle of incidence** and **angle of reflection** are equal:

$$\theta'_1 = \theta_1$$



Snell's law relates the angle of incidence and angle of refraction:

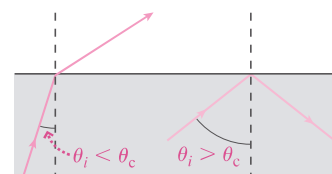
$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$



Applications

Total internal reflection results when light is incident at greater than the **critical angle**, θ_c , on an interface with a medium with lower refractive index n_2 :

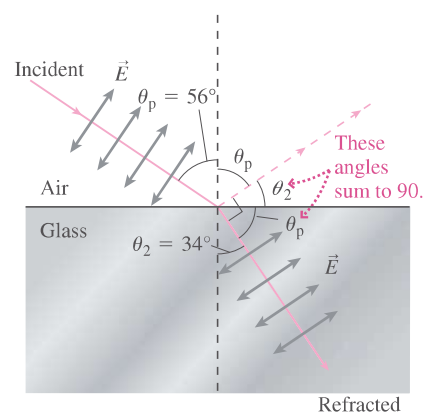
$$\sin \theta_c = \frac{n_2}{n_1}$$



Light polarized in the plane of the incident and refracted rays undergoes no reflection at an interface; this special **polarizing angle**, θ_p , is given by

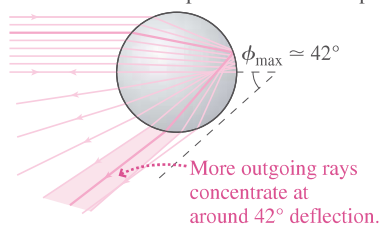
$$\tan \theta_p = \frac{n_2}{n_1}$$

For an air–glass interface, $\theta_p \approx 56^\circ$.

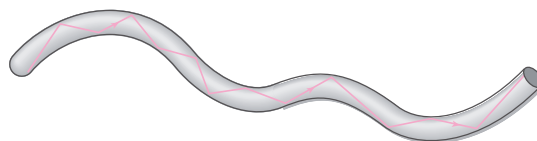


Dispersion results from the wavelength dependence of the refractive index and causes different colors to refract at different angles.

A combination of total internal reflection and dispersion in raindrops accounts for the rainbow.



Total internal reflection guides signals in optical fibers.

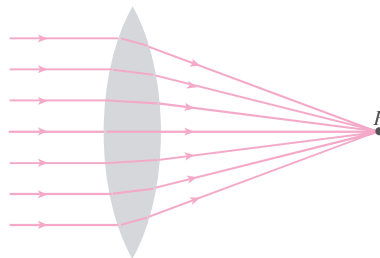
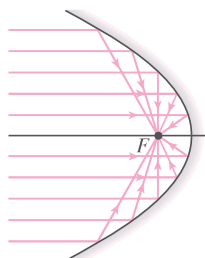


Big Picture

The big idea here is how reflection and refraction form **images**. Images are **real** or **virtual** depending on whether or not light actually comes from the image location.

Key Concepts and Equations

A curved mirror or lens has a **focal point**, F , at which parallel light rays converge:



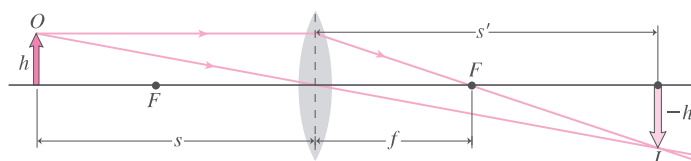
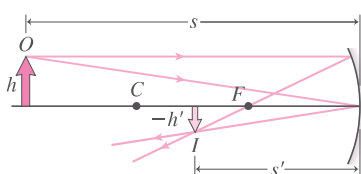
The same equation describes image formation with mirrors and lenses:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

The table summarizes the sign conventions for each term.

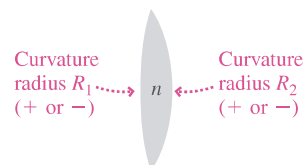
Value	Symbol	Condition	Sign
Object distance	s	Object on same side as <i>incoming</i> light rays	+
		Object on opposite side from <i>incoming</i> light rays	−
Image distance	s'	Image on same side as <i>outgoing</i> light rays	+
		Image on opposite side from <i>outgoing</i> light rays	−
Focal length	f	Focus on same side as <i>outgoing</i> light rays	+
		Focus on opposite side from <i>outgoing</i> light rays	−

Image formation with mirrors and lenses:



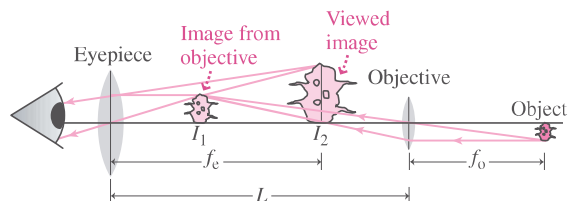
The lensmaker's formula is

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



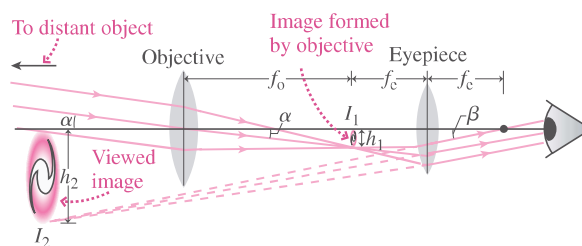
Applications

Compound microscope



Magnification: $M = -\frac{L}{f_o} \left(\frac{25 \text{ cm}}{f_e} \right)$

Refracting telescope



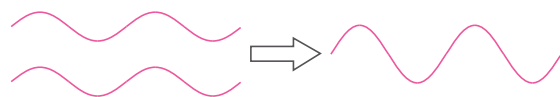
Angular magnification: $m = \frac{f_o}{f_e}$

Big Picture

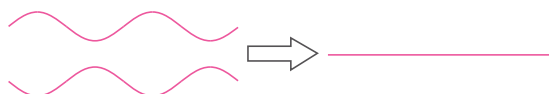
The big idea here is that—despite our use of the geometrical-optics approximation in the preceding chapters—light is indeed a wave and therefore exhibits the two related phenomena of **interference** and **diffraction**. These wave effects are important whenever light or any other wave interacts with objects whose size is comparable to or smaller than the wavelength.

Key Concepts and Equations

Constructive interference occurs when two waves combine in phase:



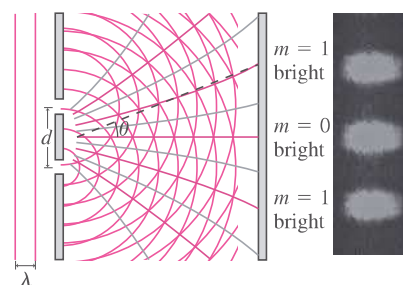
Destructive interference occurs when two waves combine 180° out of phase:



When light of wavelength λ passes through two or more narrow slits, the resulting interference shows maxima when

$$d \sin \theta = m\lambda$$

where m is an integer called the order. With multiple slits the maxima become stronger and narrower, but their position doesn't change.

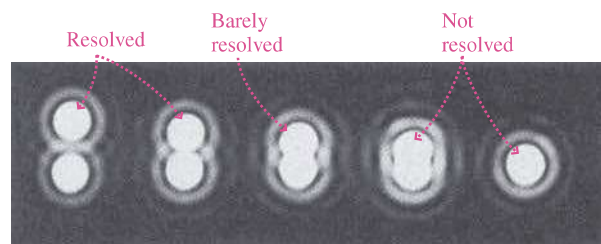


Diffraction occurs because, according to **Huygens' principle**, each point on a wavefront acts as a source of spherical waves, causing light to bend as it encounters sharp edges. Waves from different parts of a wavefront interfere to produce diffraction patterns.



The **diffraction limit** is a fundamental restriction on our ability to image small or distant objects. For a circular aperture of diameter D , the **Rayleigh criterion** gives the minimum angular separation that can be resolved with light of wavelength λ :

$$\theta_{\min} = \frac{1.22\lambda}{D}$$



Applications

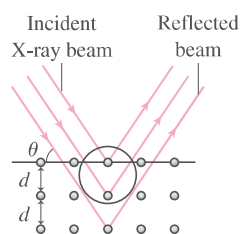
A **diffraction grating** consists of multiple slits or lines that result in constructive interference at different positions for different wavelengths. Diffraction gratings are used in spectrometers to disperse individual wavelengths. A grating's **resolving power**, the ratio of wavelength to the minimum resolvable difference in wavelengths, is given by

$$\frac{\lambda}{\Delta\lambda} = mN$$

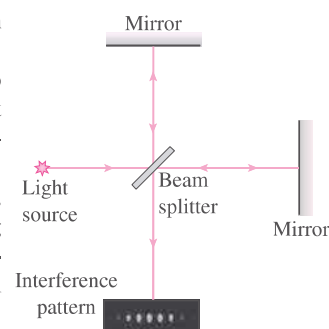
where N is the number of lines in the grating and m is the order of the dispersion.

X-ray diffraction uses regularly positioned atoms as a grating, and is a powerful technique for analyzing crystal and molecular structure. Maximum intensity occurs when

$$2d \sin \theta = m\lambda$$



The **Michelson interferometer** splits light into two beams that travel on perpendicular paths. They recombine, and the resulting interference allows precision measurements.

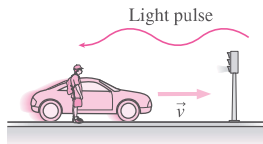


Big Picture

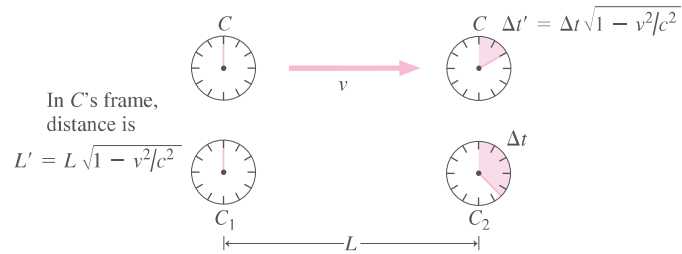
The big idea here is a simple one: The laws of physics are the same for all observers and don't depend on one's state of motion. That's the **principle of relativity**. **Special relativity** is restricted to inertial reference frames; **general relativity** removes that restriction and in so doing becomes a theory of gravity. Both relativity theories radically alter our commonsense notions of space and time. In relativity, measures of space and time—but not the laws of physics—become dependent on one's reference frame.

Key Concepts and Equations

Among the laws of physics are Maxwell's equations, with their prediction of electromagnetic waves (e.g., light) propagating at the speed of light c . Therefore, the speed of light is the same in all inertial reference frames.



Invariance of c leads to **time dilation** and **length contraction**:



Additionally, events simultaneous in one reference frame may not be simultaneous in another frame.

Time dilation and length contraction are specific instances of the **Lorentz transformations** of space and time coordinates of events observed in different reference frames. The transformations here apply to relative motion in the x -direction.

S to S'	S' to S	
$y' = y$	$y = y'$	
$z' = z$	$z = z'$	
$x' = \gamma(x - vt)$	$x = \gamma(x' + vt')$	where $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$
$t' = \gamma(t - vx/c^2)$	$t = \gamma(t' + vx'/c^2)$	

Underlying the changing measures of space and time is four-dimensional **spacetime**, in which exist **4-vectors** whose magnitude is independent of reference frame.

Invariant spacetime interval:

$$(\Delta s)^2 = c^2(\Delta t)^2 - [(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2]$$

Invariant particle mass:

$$(mc^2)^2 = E^2 - p^2c^2$$

Energy, momentum, and mass are closely related in relativity:

$$\text{Momentum: } \vec{p} = \frac{m\vec{u}}{\sqrt{1 - u^2/c^2}} = \gamma m\vec{u}$$

$$\text{Energy: } E = \frac{mc^2}{\sqrt{1 - u^2/c^2}} = \gamma mc^2 = K + mc^2 = \sqrt{(pc)^2 + (mc^2)^2}$$

Kinetic energy

Rest energy

Applications

The **Michelson–Morley experiment** of 1887 failed to detect any motion of Earth relative to the **ether**, the medium in which 19th-century physicists believed light propagated with speed c . This result helped pave the way for relativity, with no ether and with c the same in all inertial reference frames.

Relativity shows that velocities don't simply add; rather,

$$u = \frac{u' + v}{1 + u'v/c^2}$$

where u' is an object's velocity relative to a reference frame S' , u its velocity relative to frame S , and v the relative velocity between S and S' .

Einstein's famous $E = mc^2$ describes a universal interchangeability between matter and energy; contrary to common opinion, it isn't just about nuclear energy.

Big Picture

The big ideas here are at the heart of quantum physics—a radically different view of reality at the atomic scale. **Quantization** means that some physical quantities—often including energy—come only in discrete values. Another fundamental aspect of quantum reality is **wave–particle duality**, wherein light and matter exhibit both wave-like and particle-like aspects. Bohr’s **complementarity principle** precludes these ever being in direct conflict. Finally, quantization and wave–particle duality lead to the **uncertainty principle**, which states that it’s impossible to measure simultaneously and with arbitrary precision a particle’s position and momentum.

Key Concepts and Equations

Planck’s constant,

$$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$$

sets the fundamental scale of quantization.

It’s also expressed as “h bar”:

$$\hbar = h/2\pi$$

The energy of electromagnetic radiation with frequency f is quantized in **photons** with energy

$$E = hf$$

Electron energies in the **Bohr model** of hydrogen are quantized according to

$$E = -\frac{ke^2}{2a_0} \left(\frac{1}{n^2} \right) \approx -\frac{13.6 \text{ eV}}{n^2}$$

where n is an integer and $a_0 = 0.0529 \text{ nm}$ is the **Bohr radius**.

The **de Broglie wavelength** of a particle with momentum p is

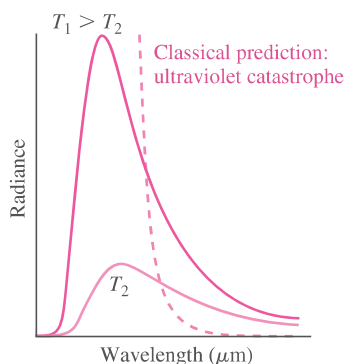
$$\lambda = \frac{h}{p}$$

The **uncertainty principle** relates uncertainties in position and momentum:

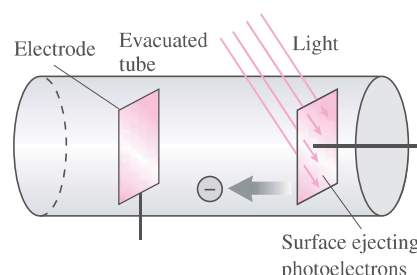
$$\Delta x \Delta p \geq \hbar$$

Applications

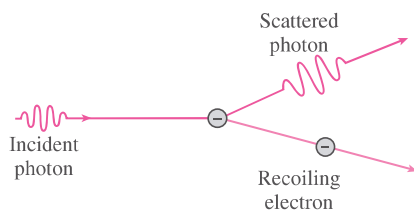
A correct description of **blackbody radiation** requires Planck’s quantization hypothesis. The peak radiance—energy radiated per unit wavelength interval—from a blackbody at temperature T occurs at a wavelength given by $\lambda T = 2.898 \text{ mm}\cdot\text{K}$.



The **photoelectric effect** involves the ejection of electrons from a metal surface illuminated with electromagnetic waves. Explanation of the effect led Einstein to propose **photons** as the quanta of electromagnetic-wave energy.



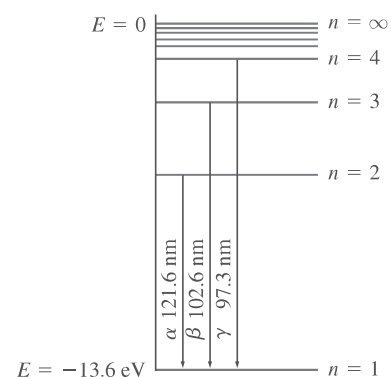
The **Compton effect** shows that photons interact with free electrons exactly like colliding particles, losing energy and emerging with longer wavelength.



Quantization of atomic energy levels leads directly to **atomic spectra**. In the Bohr model of hydrogen, the spectral line produced in a transition from the n_1 to the n_2 energy level is given by

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

where $R_H = 1.097 \times 10^7 \text{ m}^{-1}$.



Big Picture

The big idea here is the description of particles in the quantum realm using **wave functions**, whose square relates to the probability of finding a particle. Thus the link between the most thorough description physics can provide—the wave function—and the behavior of an individual particle is only statistical. The **Schrödinger equation** gives the wave function for nonrelativistic particles and leads to energy quantization for confined particles.

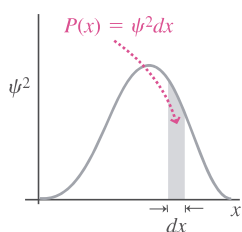
Key Concepts and Equations

The **time-independent Schrödinger equation** gives the wave function ψ for a particle of mass m with total energy E and potential energy U :

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi = E\psi$$

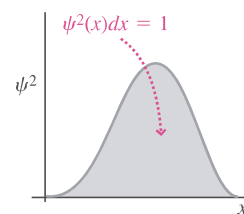
The square of the wave function is the **probability density**. In one dimension the probability of finding the particle in some small interval dx at position x is

$$P(x) = \psi^2(x) dx$$



Normalization: A particle must be somewhere, so

$$\int_{-\infty}^{+\infty} \psi^2(x) dx = 1$$



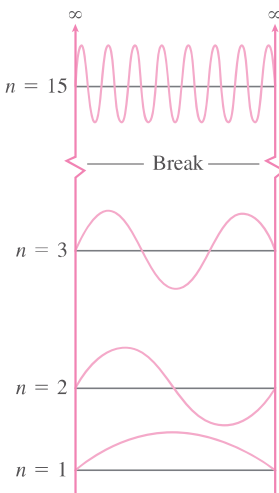
Applications

Infinite square well

Wave functions: $\psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$

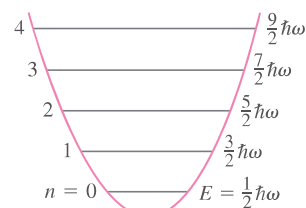
Energy levels: $E_n = \frac{n^2 h^2}{8mL^2}$

3-D well: $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$



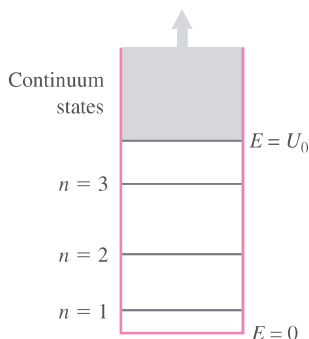
Harmonic oscillator

Energy levels: $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$



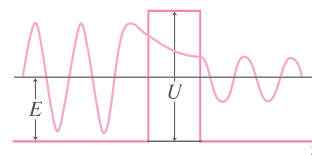
Finite well

Discrete bound states; continuum of unbound states



Quantum tunneling

Nonzero probability of finding a quantum particle in a region forbidden by classical energy conservation leads to the possibility of barrier penetration:



Big Picture

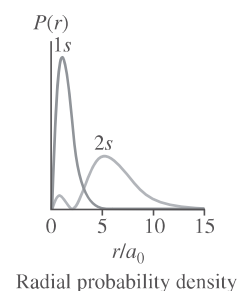
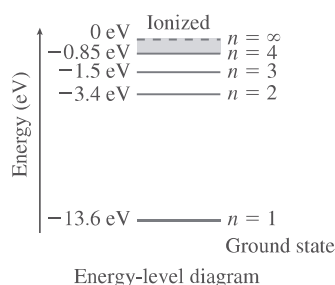
The big idea here is that atomic electrons are quantum particles trapped in the 3-dimensional potential well associated with the electric force. Solving the Schrödinger equation then leads to quantized energy levels. Considerations of electron spin and orbital angular momentum introduce subtle details into the atom's energy-level structure. The **exclusion principle** permits only one atomic electron per quantum state, and this fact underlies the shell structure of atoms and the periodic table of the elements.

Key Concepts and Equations

The **principal quantum number** n determines the energy levels in hydrogen:

$$E_n = -\frac{1}{n^2} \frac{\hbar^2}{2ma_0^2} = \frac{E_1}{n^2} = \frac{-13.6 \text{ eV}}{n^2}$$

For $n = 1$ the electron is most likely to be found one Bohr radius a_0 from the nucleus; in higher-energy states it's likely to be farther away.



The **orbital quantum number** l determines the angular momentum:

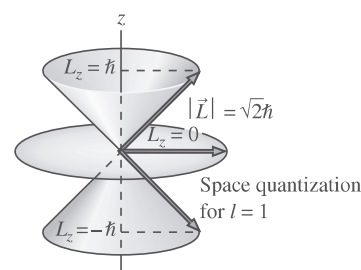
$$L = \sqrt{l(l+1)}\hbar$$

where l ranges from 0 to $n - 1$.

The **orbital magnetic quantum number** m_l determines the component of the angular momentum along any given axis:

$$L_z = m_l \hbar$$

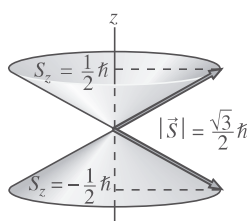
This is **space quantization**, where m_l ranges from $-l$ to l .



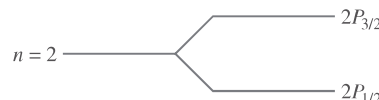
Electrons are **spin-1/2** particles or **fermions**; the component of their spin angular momentum on a given axis is $\pm \frac{1}{2}\hbar$.

Electron spin gives rise to the electron's intrinsic magnetic dipole moment, characterized by the **Bohr magneton**:

$$\mu_B = e\hbar/2m = 9.27 \times 10^{-24} \text{ A}\cdot\text{m}^2$$



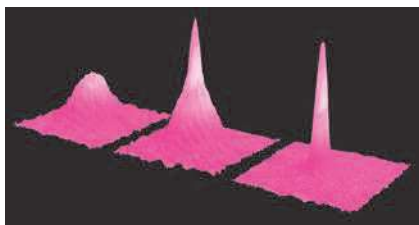
Spin-orbit coupling results in **fine-structure** splitting of atomic-energy levels.



Spin angular momentum $\vec{S}$ and total angular momentum $\vec{J}$ obey quantization rules similar to those of orbital angular momentum.

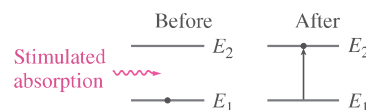
Applications

Bosons are particles with integer spin. They don't obey the exclusion principle, allowing many particles to be in the same state, as happens in a **Bose-Einstein condensate** or a laser beam.

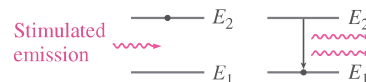


Forming a Bose-Einstein condensate

In stimulated absorption, an electron absorbs a photon and jumps to a higher energy level.



Electrons in excited states can drop to lower energy states by either **spontaneous** or **stimulated emission**.

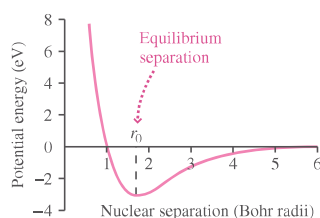


Big Picture

The big idea here is that quantum mechanics can explain the structure of molecules and solids as well as the atoms treated in Chapter 36. At this level we can't solve the Schrödinger equation for these many-particle systems, but we've argued—using energy and angular momentum quantization and the exclusion principle—that quantum effects are important in molecules and solids.

Key Concepts and Equations

Common types of molecular bonding include **ionic**, **covalent**, **hydrogen**, **van der Waals**, and **metallic** bonding. Whatever the bonding mechanism, a stable molecule is at the minimum of its potential-energy curve, shown below for H_2 .



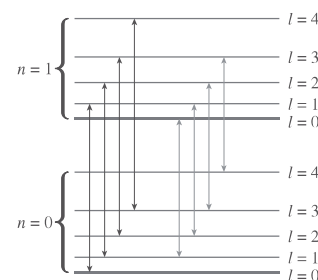
Molecules exhibit both rotational and vibrational energy, giving rise to a rich structure of quantized energy levels and the resulting spectra.

- Quantization of angular momentum leads to quantized rotational energy levels:

$$E_{\text{rot}} = \frac{\hbar^2}{2I} l(l+1), l = 0, 1, 2, \dots$$

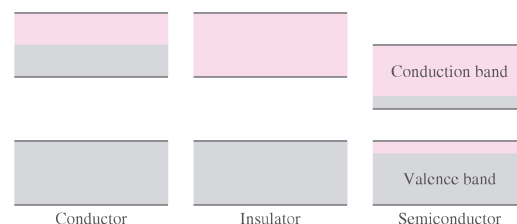
- The vibrational energy levels are those of the harmonic oscillator:

$$E_{\text{vib}} = \left(n + \frac{1}{2}\right) \hbar \omega, n = 0, 1, 2, \dots$$



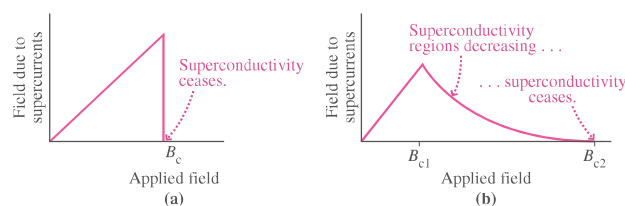
When atoms join to make solids, individual atomic energy levels separate to form **bands**. **Band theory** distinguishes conductors from insulators depending on whether the uppermost occupied band is partially or completely full, respectively. **Semiconductors** are like insulators, but with a much smaller band gap that permits thermal excitation of electrons into the conduction band.

In a metallic conductor, the energy of the highest occupied state at absolute zero is the **Fermi energy**.

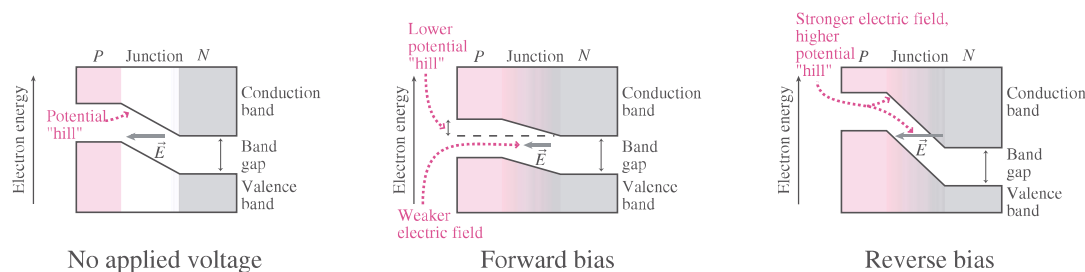


Applications

Superconductivity is a quantum-mechanical phenomenon that occurs at low temperatures and admits no classical explanation. Paired electrons move coherently through a superconductor without energy loss to the ion lattice, resulting in zero electrical resistance. Superconductors exclude magnetic fields—the **Meissner effect**—up to a **critical field** that destroys superconductivity, abruptly in **type I superconductors** and gradually in **type II superconductors**.

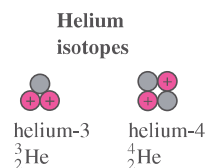


The band structure of doped semiconductors helps explain the one-way conduction of a **PN junction**, a phenomenon at the heart of modern electronics.

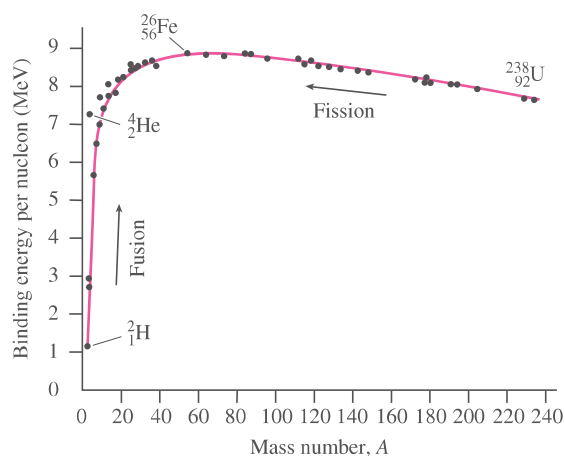


Big Picture

The big idea here is that the tiny but massive atomic nucleus is a repository of vast energy—on the order of 10^7 times the energy released in chemical reactions. The protons and neutrons that make up the nucleus can take many configurations, with the **atomic number**, Z , determining the element and the **mass number**, A , determining the particular **isotope**. We write ${}_Z^AX$ to describe the isotope with mass number A of the element whose atomic number is Z and whose symbol is X .

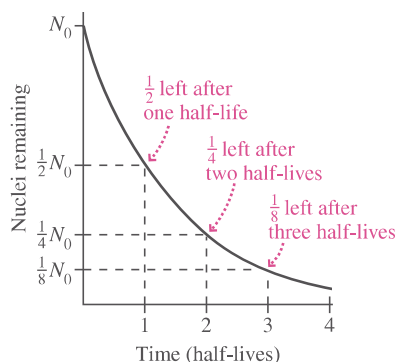


Stable isotopes require a delicate balance between protons and neutrons, with near equal numbers for lighter stable nuclei and more neutrons for heavier nuclei. **Unstable isotopes** are **radioactive** and decay by shedding particles. The **curve of binding energy** shows that energy can be released by either **fusion** of lighter nuclei or **fission** of heavier nuclei.

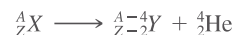


Key Concepts and Equations

Radioactive isotopes decay with a characteristic **half-life**, $t_{1/2}$: $N = N_0 2^{-t/t_{1/2}}$.



Alpha decay emits a helium-4 nucleus:



Beta decay emits an electron or a positron and an antineutrino or a neutrino:



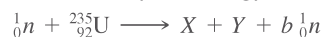
Gamma decay emits a high-energy photon (gamma ray) as an excited nucleus drops to a lower energy state:



Applications

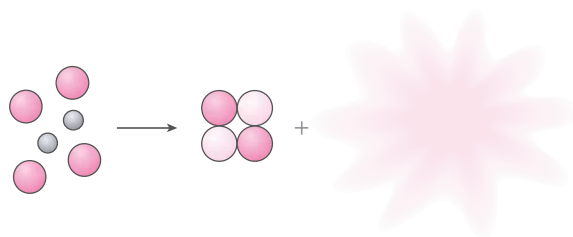
Radioactivity is measured in **becquerels**, with 1 Bq equal to one decay per second. **Sieverts** (Sv) measure the biological effects of radiation. Residents of the United States receive an average yearly radiation dose of about 3.6 mSv from both natural and artificial sources.

For fission, the most important isotopes are the **fissile** ${}_{92}^{235}\text{U}$ and ${}_{94}^{239}\text{Pu}$, which can fission when struck by low-energy neutrons:



The extra neutrons produced in fission can sustain a **chain reaction** provided there's a **critical mass** of fissile material. Exponentially growing chain reactions power fission weapons, while controlled fission occurs in **nuclear reactors** used for power generation.

Fusion powers the Sun and stars but has proved elusive on Earth except in thermonuclear weapons. **Inertial confinement** or **magnetic confinement** fusion may one day provide us with nearly limitless energy.

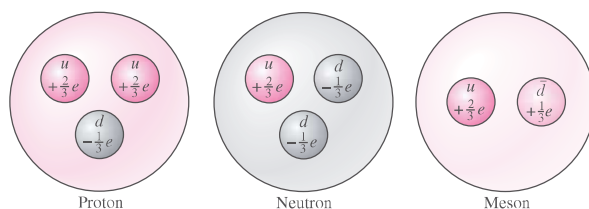


Big Picture

The big idea here is that the structure of ordinary matter and its interactions can be explained in terms of a handful of particles: **quarks**, which make up the familiar proton and neutron and a host of other **hadrons**; **leptons**, including the familiar electrons and elusive neutrinos; and **gauge bosons**, which mediate the fundamental forces and include the photon for the electromagnetic force. The **standard model** describes these elementary particles and their interactions; it sheds light not only on the structure of matter we see today but on the early universe as well. But there's a humbling caveat: Only about 4% of the universe consists of familiar matter. The rest is **dark energy** and **dark matter**, about which we know very little.

Key Concepts and Equations

Quarks, which come in six flavors and three colors, join in threes to form **baryons**, including protons and neutrons, and in twos to form **mesons**. Quarks carry fractional electric charges, but because the **strong force** between quarks doesn't decrease with distance, it appears impossible to have an isolated quark.



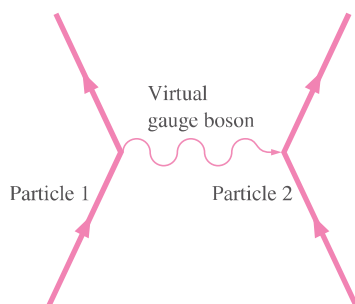
Leptons are the other class of elementary matter particles, and include the electron and its more massive cousins, the muon and tau, and the three types of neutrinos.

The three **fundamental forces**—strong, electroweak, and gravity—are believed to be manifestations of the same fundamental interactions that appear unified at high enough energies.

At typical energies in today's universe, though, the electroweak force separates into the electromagnetic and weak forces. The forces differ greatly in strength:

Force	Relative strength at 1 fm (approximate)
Gravity	10^{-38}
Weak	10^{-13}
Electromagnetic	10^{-2}
Strong	1

Exchange of **gauge bosons** explains the forces between particles.



Gauge Bosons

Electromagnetic force: photon

Weak force: $W^\pm, Z^0$

Strong force: gluon (8 varieties)

Gravity: graviton

Applications

Our knowledge of particles and their interactions, combined with general relativity and with observations of the **Hubble expansion** and the **cosmic microwave background radiation**, gives us a picture of the origin and structure of the universe. The universe began some 14 billion years ago in a hot **Big Bang**. The simplest nuclei formed within the first half-hour, and the first atoms at about 400,000 years; at that point the universe became transparent and the cosmic microwave background formed. We've learned recently that the universe is overall flat and that its expansion is accelerating under the influence of mysterious **dark energy**. There's still much that we don't know!

